Matching the Wrong Thing: Goal-Frame Velocity in the Planning Cost of a JEPA World Model

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***Abstract*- Goal observations can include state variables that a planning task does not require the agent to match. I examine this mismatch in the released JEPA-WM planner on PushT: its proprioceptive goal cost penalises the expert's goal-frame velocity, although the endpoint success criterion ignores velocity. Using the model's linear encoder, I decompose the cost and construct VELDROP-z, which retains the decoded position energy and standardises it against the visual term. The correction requires no retraining or outcome labels at inference. On a fresh cohort of 32 starts, reduced-budget cross-entropy search with VELDROP-z lowered mean executed margin from 3.057 to 1.562 (paired difference -1.495 [-2.445, -0.724]) and increased successes from 13 to 20 with no losses, repeating an earlier result on 16 goals. The pre-specified primary comparison between position and full proprioception at a shared frozen weight remained uncertain: -0.169 [-0.430, 0.091]. An additional arm, designed after analysis of the first four arms, standardised the full proprioceptive term and achieved 14 successes. VELDROP-z added 6 successes relative to this arm, but its margin difference was uncertain (-0.324 [-0.808, 0.073]). This exploratory comparison changes both the retained component and its adaptive weighting, so it does not isolate a velocity-only effect. Evaluation is limited to released checkpoints, training-split goals, a reduced search budget and open-loop execution. The evidence supports a scoped planning-cost correction and identifies the evaluations needed to establish its benefit beyond this setting.**

***Index Terms*- world models, joint-embedding predictive architecture, open-loop planning, goal-conditioned planning, cost design, reproducibility**

I. Introduction

Goal-conditioned latent world models such as DINO-WM, JEPA-WM and LeWM plan by searching for actions whose predicted terminal latent is close to the latent of a goal observation [1], [2], [3]. A growing literature asks why such planners fail although their one-step predictions are accurate, pointing to latent geometry that does not respect task costs [5], [8], [6], insensitivity to small or action-driven changes [12], [13], and prediction errors that matter for some components but not others [14].

I examine a mismatch between the goal representation and the success criterion. The general problem is well known. Goal-conditioned reinforcement learning has long defined goals on the part of the state that decides success, for example object positions without velocities [16]; full-state goals are known to over-constrain learning through nuisance coordinates such as velocities [17]; and a recent latent world model keeps velocity out of its terminal goal cost by construction [18]. I show where this mismatch enters a released planner, measure its contribution to the goal cost, and evaluate an analytic correction inside the search.

a) Contributions and scope:

1) An audit of the proprioceptive goal term of the released JEPA-WM PushT planner on its own candidates: an exact five-part decomposition (position, velocity, cross term, residual outside the encoder's range, token variation), the share of each part, its rank correlation with the decision-relevant distance, and the numerical conditioning of the decomposition (Section IV).
2) VELDROP-z, an analytic, training-free repair that keeps only the decoded position part and standardises it against the visual term (Section V).
3) Evaluations on PushT, first with fixed candidate banks and then inside CEM, including a comparison at a shared frozen weight on 32 fresh starts. DINO-WM provides a boundary case where the proposed velocity mechanism was not supported (Section VI).
4) Negative results that bound the claim: a frozen supervised scoring head that had transferred between development cohorts failed on the evaluation goals and degraded the search, and up-weighting the full proprioceptive term lost successes in fixed banks (Section VII).

This is an exploratory study of planning costs on one task. The repair does not change the world-model architecture. Restricting a goal cost to task-relevant coordinates has close precedents [16], [17], [18], [5], [6].

II. Related work

a) Planning costs for latent world models: TRM learns terminal costs on frozen latents; a decoded xy cost, or the latent distance restricted to the xy row space, repairs TwoRoom planning, and costs are combined after standardising them within the candidate set [5]. Latent Projection restricts the cost to the most dynamics-certain directions of a SIGReg latent [6]. LEAP adds a terminal-descriptor energy from a labelled decoder [7]. Learned temporal distances, manifold distances and value-shaped costs replace the Euclidean goal distance [9],

[10], [11]. AnisoWM shows that an isotropic latent induces a metric in state space that can disagree with the task cost [8]. Here I focus on a specific source of mismatch: a goal attribute that the success criterion ignores but the released planner penalises through its proprioceptive channel.

b) Criterion alignment and diagnosis: Supervising success-criterion quantities during training helps on PushT, whose release-specific criterion includes the agent position [13]. Min et al. separate representation, prediction and decision failures with matched candidates [14]; DA-LeWM measures latent-physical rank agreement along CEM stages [15]. The audit here follows the same matched-candidate logic.

c) Goal specification: Hindsight experience replay defines goals through a map from states to the coordinates that decide success, such as the object position, while the state also contains velocities [16]. GS-HER shows that full-state goals over-constrain offline goal-conditioned learners through nuisance coordinates, velocities included, and masks them with a coordinate query [17]. FIRM-WM separates a goal-comparable configuration from a dynamic state used only for prediction and excludes the latter from the terminal cost [18]. These works establish the principle and design around it; I audit released planners whose default cost includes such a coordinate and repair their cost without retraining.

III. Setting and protocol

a) Task and model: I use a reduced-budget protocol based on the released JEPA-WM PushT evaluator (repository commit `13cf1d9c`) with the released JEPA-WM checkpoint (hash-verified), on CPU [2].

- *Episodes.* An episode starts from a state of a dataset trajectory; the goal is the state reached by replaying the next 30 recorded actions.
- *Success.* The evaluator of this release (inherited from DINO-WM [1]) declares success when the Euclidean norm of the (agent xy, block xy) error is below 20 and the wrapped block-angle error is below $\pi/9$. This is the release's criterion, not a universal PushT metric [4]. I report a continuous extension, the margin $\max(\text{pos}/20,\ \text{ang}/(\pi/9))$; success holds iff it is below 1.
- *Cost and search.* The planner minimises $c = L_{\text{vis}} + 0.1\,\Pi$, where L_{vis} is the mean squared error over 16×16 DINOv2 patches of 384 channels and Π the mean squared error of the proprioceptive embedding, with the cross-entropy method (CEM) at 64 samples, 8 iterations and 10 elites (the release uses 300×30).

b) Two evaluation modes: Fixed banks: for each start, 64 plans are drawn from the final CEM Gaussian of the official-cost planner and executed; each cost selects its lowest-scoring plan, whose executed margin and success are then reported. *Inside the search:* the rule replaces the CEM objective (z-scores, where used, are computed within each 64-sample population), and the final CEM mean is executed open loop.

c) Data use, locks and decisions: Initial design used 26 starts from the validation split. The 16-goal JEPA-WM evaluation cohort and the eight-goal DINO-WM cohort use distinct trajectories from the checkpoints' *training* split: they were held out from cost design, not from checkpoint training. Later diagnostics and the in-search factorial reused the JEPA-WM goals, so I treat that sequence as exploratory development. A further cohort of 32 starts, from distinct training-split trajectories that no earlier cohort used, was drawn and locked before its runs (Section VI-C). Protocols and analysis code were hash-recorded, and the execution log records their upload to a project store before the corresponding analyses; these records are not an immutable public registration. The VELDROP-z addendum was recorded after three of 16 banks had been collected; according to the execution log, none of their outcomes had been inspected (Appendix A). A rule "helps" when its mean paired difference D is at most -0.10, its success count is not lower than the reference's, and its mean D without the most favourable start is at most zero. These are operational priority rules, not tests controlling type-I error. Intervals are percentile bootstraps over starts; exact two-sided McNemar and sign tests are descriptive. There is one keyed CEM seed per start, so planner-seed variability is not separately estimated.

IV. What the proprioceptive goal term measures

In the JEPA-WM PushT model the proprioceptive state is $r = (p, v) \in \mathbb{R}^4$, the agent's position and velocity in normalised coordinates. It is embedded by $e(r) = Wr + b$ with $W = [W_p\ W_v] \in \mathbb{R}^{16\times 4}$, and the predictor outputs $T = 256$ proprioceptive tokens. The goal is an expert frame thirty steps ahead, and its recorded velocity is usually far from zero (median speed 79 simulator units per second on the development goals).

a) Exact decomposition: For a candidate, let d_i be the difference between predicted and goal token i, $h = \frac{1}{T}\sum_i d_i$, $\delta r = W^{+}h$, $u = W_p\,\delta r_p$, $w = W_v\,\delta r_v$ and $\rho = h - W\delta r$, which is orthogonal to the range of W. Then, with $m = 16$,

$$\begin{aligned}\Pi &= \frac{1}{Tm}\sum_i \|d_i\|^2 \\ &= \frac{\|u\|^2 + \|w\|^2 + 2\langle u, w\rangle}{m} \\ &\quad + \frac{\|\rho\|^2}{m} + \frac{1}{Tm}\sum_i \|d_i - h\|^2, \end{aligned} \tag{1}$$

the five parts being position, velocity, cross term, residual and token variation. The identity follows from least-squares orthogonality and the decomposition of each token into its mean and residual. It holds even when the predicted tokens lie outside the encoder's range.

b) Measurements: On the 64-plan evaluation banks of the 16 evaluation goals (post hoc, descriptive; numerical identity error below 10^{-14}):

- mean over banks of each part's within-bank mean divided by the within-bank mean of Π: position 35.5%, velocity 62.7%, cross term 1.6%, residual and token variation together 0.17%;

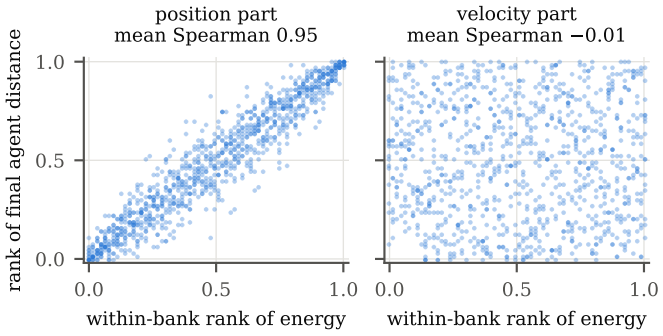


Fig. 1. Within each of the 16 evaluation banks (64 candidates of the released JEPA-WM planner), rank of the position part (left) and of the velocity part (right) of the proprioceptive goal term against the rank of the true final agent distance. The velocity part has near-zero mean within-bank rank correlation with the final agent distance; this does not imply statistical independence or absence of information (post hoc, descriptive).

- mean within-bank Spearman correlation with the true final agent distance: 0.433 for Π, 0.954 for the position part and -0.01 for the velocity part.

Velocity mismatch is thus a large part of this proprioceptive term, with near-zero *average* monotone association with the agent distance (Fig. 1). Bank-level correlations range from -0.307 to 0.347; a near-zero average neither establishes independence nor concerns every component of the success criterion. Pooling the energies of all banks gives a velocity share of 61.0%, distinct from the equal-bank average of 62.7%. On the development banks, the predicted and encoded-true values of Π have within-bank rank correlation 0.996: the ranking problem is already present when the term is evaluated on true endpoints. This does not establish that physical prediction errors are small, and I do not apportion all planning failures among prediction, objective and search.

c) Conditioning: Keeping the position part is the map $h \mapsto Ah$ with $A = W_p\,[I\ 0]\,W^+$, an *oblique* projector onto the range of W_p along the range of W_v and the orthogonal complement of the range of W. Its operator norm is $1/\sin\theta$, where θ is the smallest principal angle between the ranges of W_p and W_v, a classical fact about oblique projectors [19]. Both stored encoder matrices have full column rank. For JEPA-WM, $\theta = 59.1^\circ$ and $\|A\| = 1.17$ (condition number of W: 2.20), so the decomposition is numerically benign. Section VI-D shows that this is not so for DINO-WM.

V. The Repair

VELDROP-z scores a candidate by

$$s = z(L_{\text{vis}}) + z(\Pi_{\text{pos}}), \qquad \Pi_{\text{pos}} = \|u\|^2/m, \tag{2}$$

where $z(x) = (x - \bar{x})/(\text{sd}(x) + 10^{-12})$ uses the population standard deviation over the current candidate set. It uses no outcome labels at inference. It retains only the position energy of Eq. (1), so it removes the cross, residual and token-variation terms as well as the pure velocity term. Because $AW_v = 0$, the score is invariant to a velocity-only change of the goal embedding when the predicted tokens and the visual goal are held fixed; this algebraic invariance is not by itself a claim of better control. Its only inputs beyond the official cost are the model's proprioceptive encoder and a declared goal-coordinate schema (which entries of r are velocities).

TABLE I
FIXED-BANK SELECTION ON 16 EVALUATION GOALS (64-PLAN BANKS): MEAN SELECTED MARGIN (LOWER IS BETTER) WITH BOOTSTRAP INTERVAL, AND SUCCESSES.

Selection rule	margin [95% CI]	succ.
official cost	2.947 [1.707, 4.286]	8
VELDROP-z	**2.320** [1.360, 3.339]	**9**
supervised head (frozen)	2.909	6
proprio weight 10 (not 0.1)	2.722	5
standardised full proprio term	2.918	4
position term only	2.584	7
velocity dropped at weight 0.1	2.946	8
official cost, true endpoints	2.838	7
best plan in bank	1.966	9

I selected the rule and its weighting on the development starts, where VELDROP-z gave 3.183 with 10 successes against 3.845 and 7 for the official cost; these are design-set results, not independent estimates of performance. Development ablations also showed that dropping velocity at the official weight changes almost nothing (3.810 against 3.845), that standardising the full term loses successes (6 against 7), and that replacing the goal velocity by zero is worse than dropping it (3.275, 7 successes). The evaluations below keep the rule fixed, although earlier results informed its design. They also evaluate the frozen supervised head of an earlier development round, a ridge-regression score over the 384 visual channel energies and the proprioceptive energy fitted to development margins; it is not retrained here.

VI. Results

A. Fixed candidate banks on evaluation goals

The primary hypothesis of this locked test was the frozen supervised head against the official cost; VELDROP-z and a mechanism check were added as a separate locked hypothesis (Appendix A). Table I gives the results on 16 evaluation goals.

- *VELDROP-z improves fixed-bank selection.* $D = -0.627$ $[-1.298, -0.110]$; 7 starts improved, 1 worsened; without the most favourable start the mean is -0.371. Successes went from 8 to 9 (1 gained, 0 lost; McNemar $p = 1.000$), which equals the number of banks that contain a success: no reranking of these banks can do better on successes.
- *The position term ranks agent distance more accurately.* The position part's Spearman correlation with the agent distance (0.954) exceeds the full term's (0.43).
- *The frozen supervised head does not improve selection* ($D = -0.038$; successes 8 to 6).

a) Fixed-bank ablations (post hoc): Crossing the proprioceptive quantity (full or position) with the weight rule (fixed 0.1 or standardised) on the same banks gives margins (successes): full at 0.1, 2.947 (8); position at 0.1, 2.946 (8); full standardised, 2.918 (4); position standardised, 2.320 (9).

Neither change alone meets the operational improvement rule on these banks; their combination does. Position against full under standardisation: $D = -0.598$, successes 5 gained and 0 lost (McNemar $p = 0.0625$). Using the standardisation scale of the full term for the position term gives 2.316 (9), so in the banks the gain does not come from a different scale alone. These contrasts were computed after the outcomes were known.

B. Inside the search

I locked a second test on the same goals: CEM optimises VELDROP-z (or the frozen head) with the seeds of the official-cost CEM, and the final mean is executed. The official-cost arm is the collection's own CEM, reproduced bit-exactly by a rerun on the first start. Table II gives the results.

- *VELDROP-z lowers executed margin inside CEM.* $D = -1.468$ [-2.814, -0.296]; 13 starts improved and 3 worsened (sign test $p = 0.0213$); without the most favourable start -1.042. Successes 7 to 11, with 5 gained and 1 lost (McNemar $p = 0.219$): the success-count difference is uncertain at this sample size. The largest reduction of position error is for the agent (49.3 to 16.8; block-only successes unchanged). This is consistent with the proposed explanation but does not isolate its causal contribution.
- *The supervised head degrades in-search performance.* $D = 2.343$; successes 7 to 3 (0 gained, 4 lost). Its executed plan scored *worse under the head itself* than the official-cost plan on 7 of 16 starts (mean predicted head score 0.959 versus 0.933). Finite-budget search thus did not uniformly improve even this objective over a known feasible comparator; that observation alone does not explain the loss of executed successes or separate ranking error from optimisation.

a) Retained component and weighting inside CEM: VELDROP-z changes two things at once: it keeps only the position part, and it replaces the fixed weight 0.1 by standardisation within each population. I therefore ran the two missing cells of the 2×2 design on the same goals and seeds, under a protocol locked before these runs: the full term standardised, $z(L_{\text{vis}}) + z(\Pi)$, and the position part at the official weight, $L_{\text{vis}} + 0.1\,\Pi_{\text{pos}}$. A rerun of the official-cost search on the first start again reproduced the stored plan bit-exactly. Table III gives the four cells.

- *Standardising the full term gives the larger contrast on this cohort.* It lowered the executed margin by -1.110 [-2.328, -0.108] relative to the official cost (10 starts improved, 6 worsened) but left successes unchanged (1 gained, 1 lost). Within each population its ranking is that of $L_{\text{vis}} + \alpha_t \Pi$ with $\alpha_t = (\text{sd}(L_{\text{vis}}) + 10^{-12})/(\text{sd}(\Pi) + 10^{-12})$. Across the 128 search populations of this arm, α_t averages 9.88 (range 0.247-45.636), about 99 times the official 0.1 on average; it is not a fixed weight.
- *The standardised position-only rule has 4 more successes than the standardised full rule.* In the pre-specified primary contrast, the position part against the full term under standardisation gave $D = -0.358$ [-0.790, 0.006] (median -0.176; 10 improved, 6 worsened; without the most favourable start -0.210) and 4 additional successes with 0 lost (McNemar $p = 0.125$). The operational rule is met, but the interval reaches zero and the success difference is not significant, so this is weak evidence. The contrast changes both the retained component and its own standardisation denominator; it is not an intervention on velocity at a fixed numerical weight.
- *At the official weight, the position-only contrast is uncertain* ($D = -0.075$ [-0.437, 0.294]; successes 1 gained, 1 lost), as in the fixed banks. This is consistent with the scales logged in this arm: at weight 0.1, the position term's standard deviation within a search population is on average 91 times smaller than the visual term's, and at least 5.8 times smaller in every population. Interaction -0.283 [-0.760, 0.181].

The path from full at 0.1, to standardised full, to standardised position has margin changes -1.110 and -0.358, summing to -1.468; the path through position at 0.1 has changes -0.075 and -1.393. Neither path is a causal allocation of the total gain. The same holds for successes: the four extra successes of VELDROP-z appear in the last step of either path, so they cannot be attributed to velocity removal alone. Isolating the retained component requires a comparison at one weight frozen from a shared initial candidate population; Section VI-C reports one, and repeats this factorial contrast, on a fresh cohort. Finally, the fixed-bank experiment selects among raw-planner candidates, whereas the in-search experiment changes those candidates and executes an elite mean, so different outcomes in the two modes need not conflict.

C. A fresh cohort

To repeat the in-search tests on goals that played no earlier role, I locked a test on 32 new starts. Each start runs CEM arms with the same seed, so all arms share the first population Q_0: the official cost, VELDROP-z, $L_{\text{vis}} + \alpha_0 \Pi$ and $L_{\text{vis}} + \alpha_0 \Pi_{\text{pos}}$. Here $\alpha_0 = (\text{sd}(L_{\text{vis}}) + 10^{-12})/(\text{sd}(\Pi) + 10^{-12})$ is computed once on Q_0 and frozen for all iterations; it averages 1.64 (range 0.34-4.07), about 16 times the official 0.1. The primary question was the frozen-weight comparison of the two terms; VELDROP-z against the official cost was a pre-declared secondary question. After these four arms had been analysed, a fifth arm, the standardised full term $z(L_{\text{vis}}) + z(\Pi)$, was locked and run on the same starts to repeat the primary contrast of Table III. Table IV gives all five arms.

- *VELDROP-z against the official cost (pre-specified secondary).* $D = -1.495$ [-2.445, -0.724] (median -0.539; 26 starts improved, 6 worsened; sign test $p = 0.0005$; without the most favourable start -1.189). Successes rose from 13 to 20, with 7 gained and 0 lost (exact McNemar $p = 0.0156$). This repeats the earlier in-search result on fresh goals. The largest mean error reduction is again for agent placement (agent error 52.9 to 16.5), while the mean block error rose from 19.3 to 23.4.
- *Position against full at the frozen weight (primary).* $D = -0.169$ [-0.430, 0.091] (median -0.136; 22 improved,

TABLE II

THE SCORE USED AS THE CEM OBJECTIVE, 16 GOALS (SAME GOALS AS TABLE I); FINAL CEM MEAN EXECUTED OPEN LOOP. MEAN EXECUTED MARGIN WITH BOOTSTRAP INTERVAL, SUCCESSES, BLOCK-ONLY SUCCESSES AND MEAN ERRORS.

CEM objective	margin [95% CI]	succ.	block only	agent err.	block err.
official cost	3.059 [1.698, 4.476]	7	12	49.3	26.1
VELDROP-z	**1.591** [0.817, 2.536]	**11**	12	16.8	22.9
supervised head (frozen)	5.402 [3.058, 8.228]	3	4	83.7	51.0

TABLE III

THE 2×2 FACTORIAL INSIDE CEM ON THE 16 GOALS OF TABLE II: MEAN EXECUTED MARGIN (LOWER IS BETTER), SUCCESSES AND MEAN FINAL AGENT ERROR. THE FULL TERM AND THE POSITION PART AT WEIGHT 0.1 VERSUS STANDARDISED WITHIN EACH POPULATION.

proprio quantity, weight	margin	succ.	agent err.
full, 0.1 (official)	3.059	7	49.3
position, 0.1	2.983	7	47.4
full, standardised	1.949	7	24.4
position, standardised (VELDROP-z)	1.591	11	16.8

TABLE IV

FRESH COHORT OF 32 STARTS; THE SCORE IS THE CEM OBJECTIVE AND THE FINAL MEAN IS EXECUTED OPEN LOOP. MEAN EXECUTED MARGIN, SUCCESSES, BLOCK-ONLY SUCCESSES, AND MEAN AGENT AND BLOCK ERRORS.

CEM objective	margin	succ.	block only	agent	block
official cost	3.057	13	21	52.9	19.3
full, frozen α_0	2.597	14	23	43.9	18.9
position, frozen α_0	2.429	14	23	39.9	17.4
full, standardised	1.887	14	23	23.3	22.1
VELDROP-z	**1.562**	**20**	22	16.5	23.4

10 worsened; sign test $p = 0.0501$; without the most favourable start -0.087), with 1 success gained and 1 lost. The operational rule is met, but the interval contains zero. The point estimate favours the position term; the interval also permits a larger gain or a slight degradation. This is not an equivalence test. The contrast removes the cross, residual and token-variation terms together with the velocity term.

- *Descriptive contrasts.* Both frozen-weight rules improve on the official cost (full -0.460 [-0.857, -0.145], position -0.629 [-0.991, -0.328]), with one net success each. VELDROP-z improves on the frozen-weight position rule by -0.867 [-1.590, -0.294], with 6 successes gained and 0 lost.
- *Position against full under standardisation (exploratory added arm).* $D = -0.324$ [-0.808, 0.073] (median -0.183; 21 improved, 11 worsened; sign test $p = 0.1102$; without the most favourable start -0.165), with 6 successes gained and 0 lost (exact McNemar $p = 0.0312$). Standardising the full term itself changes the margin by -1.171 [-2.024, -0.520] relative to the official cost, with 2 successes gained and 1 lost. Of the 7 starts that VELDROP-z adds over the official cost, two are already successes under the standardised full term.

On the new cohort, the path from the official cost to standardised full and then to standardised position has margin changes -1.171 and -0.324, respectively. The latter comparison has 6 success gains and 0 losses, but its margin interval includes zero. It changes both the retained term and its adaptive weighting. At the frozen intermediate weight, the estimated margin difference is uncertain and the success count is unchanged. The weight that VELDROP-z implicitly gives the position term, $(\mathrm{sd}(L_{\mathrm{vis}}) + 10^{-12})/(\mathrm{sd}(\Pi_{\mathrm{pos}}) + 10^{-12})$, has a median of 3.74 in the first population and 23.47 in the last (post hoc), whereas α_0 stays at its initial value. One reading, not tested here, is that the velocity-related terms matter mainly when proprioception carries a large weight. As before, the standardised contrast also changes the standardisation denominator, so no path through these arms is a causal allocation of the total gain.

D. A second released model

DINO-WM uses the same frozen DINOv2 encoder, its own predictor, a $\mathrm{Linear}(4 \to 20)$ proprioceptive encoder, and a goal cost of the same form. I locked an unchanged VELDROP-z test on 8 new training-split goals (fixed banks).

- *Selection improves slightly, with unchanged successes.* 3.506 against 3.862 ($D = -0.355$; 3 improved, 0 worsened, 5 unchanged; sign test $p = 0.25$). With five zero differences, a bootstrap resample consists only of zeros with probability 0.0233, so the percentile interval is not strong evidence. The official cost and both standardised rules attain the same 2 successful banks.
- *Velocity mismatch is small in DINO-WM.* Velocity carries 0.6% of the proprioceptive term (position 98.5%), and the full term already ranks candidates by the agent distance (Spearman 0.980).

DINO-WM's encoder is much less sensitive to velocity in normalised coordinates: its velocity columns have norms 0.021 and 0.019, against 0.88 and 0.73 for position. They are also nearly parallel to the position columns ($\theta = 1.0°$, condition number 2562.2), so the position-only map amplifies embedding errors up to $\|A\| = 56.64$ times. On these banks VELDROP-z selected the same plan as the standardised full term on 8 of 8 starts: re-weighting the full term reproduces the observed selections, so these banks show no incremental selection benefit from dropping the velocity-related terms. Because the stored predictions are rounded to 16 bits, I re-scored every DINO-WM evaluation bank in 32 bits (and four JEPA-WM banks as a control). The new inference reproduced the stored official costs exactly and, rounded to 16 bits, the stored tokens bit for bit; no selection of VELDROP-z, of

the standardised full term or of the official cost changed (8 DINO-WM and 4 JEPA-WM banks), and the largest relative change of the position energy was 5.4% (DINO-WM) and 0.4% (JEPA-WM). The precision check leaves the selections unchanged on these banks. The DINO-WM projection remains poorly conditioned for perturbations in other directions.

VII. Negative controls

The frozen supervised scoring head had shown a favourable transfer estimate in earlier development, yet it produced 6 successes in the JEPA-WM fixed banks against 8 for the official cost, and 3 against 7 inside CEM. Increasing the full proprioceptive weight from 0.1 to 10 (100-fold), or standardising the full term, lowered the mean margin but lost successes in fixed banks (Table I); inside CEM, standardising the full term lowered the margin while keeping the success count at 7 (1 gain, 1 loss). Aggregate rank correlations and fixed-bank outcomes therefore cannot substitute for evaluating a cost inside the search. Two training-free visual energies from earlier rounds also failed their own locked tests: an optimal-transport patch energy was refuted by its mechanism probe, and tail pooling of patch errors gave $D = -0.053$ [-0.826, 0.797]. I report these failures alongside the positive rule.

VIII. Limitations

- *Scope.* One task; two released checkpoints; CEM at 64×8 rather than 300×30; one open-loop plan rather than receding-horizon control.
- *Dependence and statistics.* The fixed-bank, in-search and factorial tests use the same 16 goals and are not independent replications; the fresh cohort of 32 starts is the only repetition on new goals, and the DINO-WM test is a fixed-bank test. Several decisions were pre-specified across rounds and earlier development rounds chose the direction, so the results are not corrected for multiplicity. On the fresh cohort, two questions were pre-declared and a third was added after their analysis; its success differences have exact McNemar values of 0.0156 (VELDROP-z against the official cost) and 0.0312 (position against full under standardisation), uncorrected. Other tested success-count differences did not reach the nominal 0.05 threshold.
- *Data.* All evaluation goals, the fresh cohort included, come from training-split trajectories of the checkpoints.
- *Mechanism scope.* The mechanism needs a goal whose proprioception contains a velocity the criterion ignores and an encoder that is sensitive to it. The JEPA-WM banks show a large velocity-energy contribution and the DINO-WM banks a much smaller one, although DINO-WM's encoder also has nonzero velocity columns and full rank. Other tasks are untested.
- *Design.* The equal-weight standardisation is a heuristic, robust over a range of weights in development. The goal-coordinate schema is task knowledge.

IX. Conclusion

The released JEPA-WM PushT cost penalises a goal-frame velocity that its endpoint success criterion does not require. An exact linear decomposition exposes a large velocity-energy contribution whose mean within-bank rank correlation with the final agent distance is near zero. A position-only, standardised cost (VELDROP-z) improved the executed plans of reduced-budget search, first on 16 goals and then on a fresh cohort of 32 starts, where successes rose from 13 to 20 with no losses. On both cohorts, the official-cost to standardised-full contrast is larger in mean margin than the standardised-full to standardised-position contrast. The latter comparison has four and six success gains, respectively, with no losses, but both margin intervals include zero. At a frozen intermediate weight the estimated difference remains uncertain. These comparisons do not identify a pure-velocity effect or allocate the total gain causally. DINO-WM is a boundary case with no incremental selection benefit over standardised full proprioception. A practical audit for any goal-conditioned latent planner follows: decompose the goal term by what the goal encodes, compare each part with what the success criterion depends on, and test a repair inside the search rather than only on fixed candidates. Full-budget receding-horizon evaluation, several seeds, goals outside the training split and other tasks are needed before broader claims.

Appendix A
Locks, timestamps and deviations

- *Fixed-bank test.* Locked at 02:32 UTC on 27 Oct 2026 before any data. The VELDROP-z addendum, the mechanism check and the in-search protocol were locked at 02:52, when 3 of 16 starts had been collected; according to the execution log, none had been opened.
- *DINO-WM test.* Locked at 04:58 and amended at 05:00 to a fixed eight-start cohort, before any data.
- *Factorial inside the search.* Designed after the earlier outcomes were known, and recorded at 23:48 UTC on 27 Oct 2026 (06:48 on 28 Oct, Vietnam time) before its runs; it is development evidence on the same goals.
- *Fresh cohort.* Locked at 07:54 UTC on 28 Oct 2026, after infrastructure checks on an old start (the official-cost search reproduced its stored plan exactly) and before any run of the cohort. It ran in nine resumable chunks whose logs print timing and α_0 only, and it was analysed once, after all 128 runs, then re-checked by a second implementation. No run crashed.
- *Standardised full arm on the fresh cohort.* Added after the analysis of the first four arms and locked at 23:41 UTC on 28 Oct 2026 before its runs. A restart of the computing container stopped its collection after four runs; the interrupted run was repeated from its seed with unchanged code, and the interruption is recorded in the deviation log. It was analysed once and re-checked by a second implementation.
- *Execution deviations.* The in-search test ran in three resumable chunks; the console log of the first was overwritten

by a relaunch (each run record keeps its own timing), and the second recorded nothing. While the results file was inspected for its format, two individual run records were printed before the analysis. A first DINO-WM launch, started concurrently with the in-search test contrary to its lock, was killed before it recorded any start.

- *Post hoc material.* The five-part decomposition shares, the bank factorial, the encoder geometry, the agreement between VELDROP-z and the standardised full term on DINO-WM, and the head's own-score comparison were computed after the outcomes and are descriptive.
- *Verification.* Separate analysis implementations reproduced the recorded numerical decisions from saved per-plan components and terminal states. The fresh-cohort audit additionally checks all 160 terminal records, five-arm goal and initial-population matching, and constancy of the frozen weights. This verifies the saved-result analysis, not an independent repetition of model training or of all simulator collection. The accompanying artifact provides portable saved-result analyses and external-runtime collection scripts. A complete collection rerun requires the released checkpoints, datasets and pinned upstream repositories; those dependencies are not bundled. The analysis code and saved outputs are prepared for release through the GitHub page listed with the author information.

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